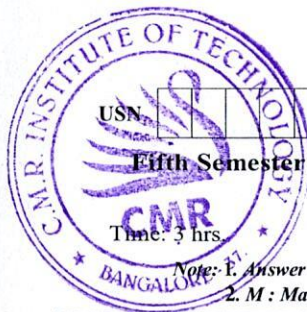




# CBCS SCHEME

BCS515B

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

## Artificial Intelligence

Time: 3 hrs

Max. Marks: 100

Note: I. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks, L: Bloom's level, C: Course outcomes.

| Module – 1 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   | M  | L   | C |
|------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|---|
| Q.1        | a. | Define the following :<br>i) Intelligence ii) Artificial Intelligence iii) Agent iv) Rationality<br>v) Logical reasoning.                                                                                                                                                                                                                                                                                                                                                   | 5 | L2 | CO1 |   |
|            | b. | Examine the AI literature to discover whether the following tasks can currently be solved by computers.<br>i) Playing a decent game of table tennis (ping-pong)<br>ii) Discovering and proving new mathematical theorems<br>iii) Giving competent legal advice in a specialized area of law<br>iv) Performing a complex a surgical operation.                                                                                                                               | 8 | L2 | CO1 |   |
|            | c. | Implement a simple reflex agent for the vacuum environment. Run the environment with this agent for all possible initial dirt configurations and agent locations. Record the performance score for each configuration and the overall score.                                                                                                                                                                                                                                | 7 | L3 | CO1 |   |
| OR         |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |    |     |   |
| Q.2        | a. | Is AI a science, or is it engineering or neither or both? Explain.                                                                                                                                                                                                                                                                                                                                                                                                          | 5 | L2 | CO1 |   |
|            | b. | Write pseudocode agent programs for the goal based and utility based agents.                                                                                                                                                                                                                                                                                                                                                                                                | 8 | L1 | CO1 |   |
|            | c. | For each the following activities give a PEAS description.<br>i) Playing a tennis match<br>ii) Performing a high jump<br>iii) Bidding on an item in an auction.                                                                                                                                                                                                                                                                                                             | 7 | L1 | CO1 |   |
| Module – 2 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |    |     |   |
| Q.3        | a. | Explain why problem formulation must follow goal transformation.                                                                                                                                                                                                                                                                                                                                                                                                            | 5 | L1 | CO1 |   |
|            | b. | Give complete problem formulation for each of the following choose a formulation that is precise enough to be implemented.<br>i) Using only four colors, you have to color a planar graph in such a way that no two adjacent regions have the same color.<br>ii) A 3 – foot – tall monkey is in a room where some bananas are suspended from the 8-foot ceiling. He would like to get the bananas. The room contains two stackable, moveable, climbable 3-foot high crates. | 8 | L2 | CO2 |   |
|            | c. | Prove each of the following statements or given counter example :<br>i) Breadth – first search is a special case of uniform – cost search.<br>ii) Uniform – cost search is a special case of A* search.                                                                                                                                                                                                                                                                     | 7 | L2 | CO2 |   |

| OR         |    |                                                                                                                                                                                                                                                                                                                                                                                    |   |    |     |  |
|------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|--|
| Q.4        | a. | Define the following terms with example.<br>i) State space ii) Search node iii) Transition model<br>iv) Branching factor.                                                                                                                                                                                                                                                          | 8 | L2 | CO2 |  |
|            | b. | Show that the 8-puzzle states are divided in to two disjoint sets, such that any state is reachable from any other state in the same set, while no state is reachable from any state in the other set. Devise a procedure to decide which set a given state is in and explain why this is useful for generating random state.                                                      | 7 | L2 | CO2 |  |
|            | c. | Describe a state space in which iterative deepening search performs much worse than depth first search for example, $O(n^2) \forall O(n)$ .                                                                                                                                                                                                                                        | 5 | L2 | CO2 |  |
| Module – 3 |    |                                                                                                                                                                                                                                                                                                                                                                                    |   |    |     |  |
| Q.5        | a. | Devise a state space in which A* using GRAPH-SEARCH returns a suboptimal solution with h(n) function that is admissible but inconsistent.                                                                                                                                                                                                                                          | 7 | L2 | CO3 |  |
|            | b. | Which of the following are correct?<br>i) $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \vee (A \vee B)$<br>ii) $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \vee (A \vee B) \wedge (\neg D \vee E)$<br>iii) $(A \vee B) \wedge \neg(A \Rightarrow B)$ is satisfiable<br>iv) $(A \Leftrightarrow B) \Leftrightarrow C$ has the same number of models as $(A \Leftrightarrow B)$ | 8 | L1 | CO3 |  |
|            | c. | Consider a vocabulary with only four propositions, A, B, C and D. How many models are there for the following sentences?<br>i) $B \vee C$ ii) $\neg A \vee \neg B \vee \neg C \vee \neg D$ iii) $(A \Rightarrow B) \wedge A \wedge \neg B \wedge C \wedge D$ .                                                                                                                     | 5 | L1 | CO3 |  |
| OR         |    |                                                                                                                                                                                                                                                                                                                                                                                    |   |    |     |  |
| Q.6        | a. | Prove that if a heuristic is consistent, it must be admissible. Construct an admissible heuristic that is not consistent.                                                                                                                                                                                                                                                          | 8 | L1 | CO3 |  |
|            | b. | Prove each of the following assertions :<br>i) $\alpha \equiv \beta$ if and only if the sentence $(\alpha \Leftrightarrow \beta)$ is valid<br>ii) $\alpha \neq \beta$ if and only if the sentence $\alpha \wedge \neg \beta$ is unsatisfiable.                                                                                                                                     | 7 | L1 | CO3 |  |
|            | c. | Prove, or find a counter example to each of the following assertions.<br>i) If $\alpha \neq (\beta \wedge \gamma)$ then $\alpha \neq \beta$ and $\alpha \neq \gamma$<br>ii) If $\alpha \neq (\beta \vee \gamma)$ then $\alpha \neq \beta$ and $\alpha \neq \gamma$ (or) both                                                                                                       | 5 | L1 | CO3 |  |
| Module – 4 |    |                                                                                                                                                                                                                                                                                                                                                                                    |   |    |     |  |
| Q.7        | a. | Which of the following are valid necessary true sentences?<br>i) $(\exists x x = x) \Rightarrow (\forall y \exists z y = z)$<br>ii) $\forall x P(x) \vee \neg p(x)$<br>iii) $\forall x \text{ smart}(x) \vee (x = x)$                                                                                                                                                              | 7 | L1 | CO4 |  |
|            | b. | Prove that universal Instantiation is sound that existential instantiation produces an inferentially equivalent knowledge base.                                                                                                                                                                                                                                                    | 5 | L1 | CO4 |  |

|                   |    |                                                                                                                                                                                                                                                                                                                                                                           |   |    |     |
|-------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
|                   | c. | Write down logical representations for the following sentences, suitable for use with generalized modulus ponens :<br>i) Horses, cows and pigs are mammals<br>ii) Bluebeard is Charlie's parent<br>iii) Offspring and parent are inverse relations                                                                                                                        | 8 | L1 | CO4 |
| <b>OR</b>         |    |                                                                                                                                                                                                                                                                                                                                                                           |   |    |     |
| Q.8               | a. | Consider a knowledge base containing just two sentence ; P(a) and P(b). does this knowledge base entail $\forall x P(x)$ ? Explain your answer in terms of models.                                                                                                                                                                                                        | 5 | L2 | CO4 |
|                   | b. | Suppose a knowledge base contains just one sentence, $\exists x \text{AsHighAs}(x, \text{Everest})$ which of the following are legitimate results of applying existential instantiation?<br>i) $\text{AsHighAs}(\text{Kilimanjaro}, \text{Everest})$<br>ii) $\text{AsHighAs}(\text{Kilimanjaro}, \text{Everest}) \wedge \text{AsHighAs}(\text{Benvevis}, \text{Everest})$ | 8 | L2 | CO4 |
|                   | c. | Explain how to write any 3-SAT problem of arbitrary size using a single first order definite clause and no more than 30 ground facts.                                                                                                                                                                                                                                     | 7 | L2 | CO4 |
| <b>Module – 5</b> |    |                                                                                                                                                                                                                                                                                                                                                                           |   |    |     |
| Q.9               | a. | i) Give a backward chaining proof of the sentence $7 \leq 3 + 9$ . Show only the steps that leads to success<br>ii) Give a forward chaining proof of the sentence $7 \leq 3 + 9$ . Show only the steps that leads to success.                                                                                                                                             | 8 | L1 | CO5 |
|                   | b. | Describe the differences and similarities between problem solving and planning.                                                                                                                                                                                                                                                                                           | 5 | L2 | CO5 |
|                   | c. | Prove that backward search with PDDL problems is complete.                                                                                                                                                                                                                                                                                                                | 7 | L1 | CO5 |
| <b>OR</b>         |    |                                                                                                                                                                                                                                                                                                                                                                           |   |    |     |
| Q.10              | a. | The following prolog code defines a predicate P<br>$P(x, [x y])$ ,<br>$P(x, [y z]) :- P(x, z)$<br>i) Show proof trees and solutions for the queries $P(A, [2, 1, 3])$ and $P(z, [1, A, 3])$<br>ii) What standard list operation does P represent?                                                                                                                         | 8 | L1 | CO5 |
|                   | b. | Explain why dropping negative effects from every action schema in a planning problem results in a relaxed problems.                                                                                                                                                                                                                                                       | 5 | L2 | CO5 |
|                   | c. | Prove the following assertions about planning graphs :<br>i) A literal that does not appear in the final level of the graph cannot be achieved.<br>ii) The level cost of a literal in a serial graph is no greater than the actual cost of an optimal plan for achieving it.                                                                                              | 7 | L1 | CO5 |

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